BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 202 · Boolean identities and De Morgan's laws

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Using h(k)=k MOD 5, what is the starting bucket for 17?

Q2. What does an adjacency list store for a vertex?

Q3. What is the result of TRUE AND FALSE?

Q4. Simplify A ∧ 1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Boolean identity — Rule for simplifying logical expressions.

De Morgan's law — Negate each operand and swap AND and OR.

complement — Logical NOT of a value or expression.

double negation — ¬¬A simplifies to A.

equivalence — Two expressions give the same output for every combination of inputs.

New learning

Negate the inputs — Distribute NOT across the brackets by negating A and B separately.

Swap the operator — When applying De Morgan's law, AND changes to OR; OR changes to AND.

Check equivalence — Both expressions produce the same output for all four combinations of A and B.

Worked example

Simplify ¬(A ∨ B) and verify its truth table.

1. Predict and explain — Negate A and B separately and replace OR with AND.

2. Check and extend — Both expressions are true only when A=0 and B=0, proving equivalence.

We Do · Practise with guidance

Simplify ¬(A ∧ ¬B) without drawing a gate diagram.

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¬(A ∧ ¬B) = ¬A ∨ ¬¬B = ?
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Predict and explain

De Morgan gives ¬A ∨ ¬(¬B).

Check and extend

Double negation gives ¬A ∨ B. The two expressions have matching outputs.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Simplify A ∨ 0.

Question 2

Use De Morgan to simplify ¬(P ∧ Q).

Question 3

Simplify ¬(A ∨ ¬B) and justify the final result.

Exit Ticket · Answer before checking

De Morgan's law allows us to remove a NOT from brackets without changing AND or OR.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 2.

Q2. The neighbouring vertices.

Q3. FALSE.

Q4. A. AND with true leaves A unchanged.

You Do pages · Independent questions

Q1. A. OR with false does not change the result.

Q2. ¬P ∨ ¬Q. Negate each input and change AND to OR.

Q3. ¬A ∧ B. De Morgan gives ¬A ∧ ¬¬B; double negation yields B.

Exit Ticket · Model answer

False. AND and OR must swap when the NOT is distributed.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: